DO NOW — RECALL

Karnaugh maps 1: start with four squares

Name: _____ Class: _____ Date: _____

1. Write −13 as an 8-bit two’s-complement integer .

2. For the circuit shown, simplify Q = ¬(A ∨ B) using De Morgan’s Law.

3. Complete one full left-to-right pass of ascending bubble sort. What is the resulting array?

4. Normalise this positive binary mantissa without changing its value. Give the new exponent.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

11110011 . Start with +13 = 00001101. Invert the bits → 11110010, then add 1 → 11110011.

2. Model answer

Q = ¬A ∧ ¬B . NOT of an OR becomes AND of the individually negated inputs. Both expressions are equivalent.

3. Model answer

[4, 6, 2, 8, 9] . Compare neighbouring items and swap where needed: [4,9,6,2,8] → [4,6,9,2,8] → [4,6,2,9,8] → [4,6,2,8,9].

4. Model answer

0.110100×2^{2} . Shift the mantissa point two places right (0.001101 → 0.110100), so decrease the exponent from 4 to 2. The numerical value stays the same.

By the end, you can…

- transfer a two-input truth table into a 2×2 map;
- group adjacent 1s and explain why changing inputs are removed;
- explain why three-input maps use Gray-code columns 00, 01, 11, 10;
- simplify groups by keeping fixed inputs and removing changing ones.

Vocabulary — know these terms

Recall the term — Karnaugh map

Recall the term — Group

Recall the term — Adjacent

Recall the term — Simplify

Recall the term — Gray code

NEW LEARNING

Read, annotate and explain

Truth table → four-square map

Each cell tells us which inputs produced the output. In a group, keep the inputs that do not change. Optional explanatory video ↗

Map the truth-table outputs

For each combination of A and B, find the same output Q in the map. The grid moves outputs; it does not change them.

Group adjacent 1s

Circle the two 1s in the bottom row. A group can contain 1, 2, 4, 8… adjacent 1s. Use the largest useful group.

Compare the input values

For the two circled cells:

A stays 1; B changes from 0 to 1.

Keep only what stays the same

Look at each input variable across the grouped cells (where Q = 1):

In our group, A = 1 in both cells, while B changes. Keep A, leave out B: Q = A .

Why Gray code?

From 01 to 10 , two bits change — these cannot be neighbouring K-map columns.

Each neighbouring pair differs by one bit only , including the end pair 10 ↔ 00 .

Read the three-input layout

Two bits label each BC column. Moving sideways changes one input only. This allows useful groups in three-variable maps.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

How does this four-square map simplify?

Read the input combinations

Only the two bottom cells contain 1. They represent A=1, B=0 and A=1, B=1 .

Circle the adjacent 1s

Both outputs are 1 and the squares share an edge.

Compare each input

Write the simplified expression

Keep A because it remains 1. Remove B because it changes between 0 and 1.

Check: Q is 1 for both values of B whenever A = 1.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Circle a vertical pair

Two 1s lie in the same column.

Method / working:

2. Circle a horizontal pair

Two 1s lie in the same row.

Method / working:

3. Find two useful pairs

All three 1s must be covered.

Method / working:

4. First three-input simplification

Column labels are BC. Only neighbouring labels differ by one bit.

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

1. One horizontal pair
Circle both adjacent 1s. For A and B, state which stays constant or changes. Write the simplified Q.

Question 2

2. Just one 1
Which input values produce the only 1? Write Q as one AND term using A and B.

Question 3

3. Two pairs with an overlap
Circle two useful pairs, identify the fixed input in each, then combine the terms with OR.

Question 4

4. Explain Gray-code order
Complete the two missing column labels, then explain why 01 and 11 are adjacent and why the order 00, 01, 10, 11 is unsuitable for a K-map.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 5

5. Simplify a group of four
The columns use BC Gray-code order. Circle all four 1s, compare A/B/C, then write Q.

Question 6

6. Simplify a pair
Circle the two adjacent 1s. Which two inputs stay constant? Write the simplified Boolean term.

Question 7

7. Free-popcorn voucher
E = voucher shown, F = soft drink, G = water. Free popcorn is offered when E AND (F OR G). Circle two useful pairs and write R as an OR of two AND terms. Adapted from OCR Boolean Algebra Q2(b).

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A variable that changes value inside a Karnaugh-map group is removed from that group's simplified term.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.